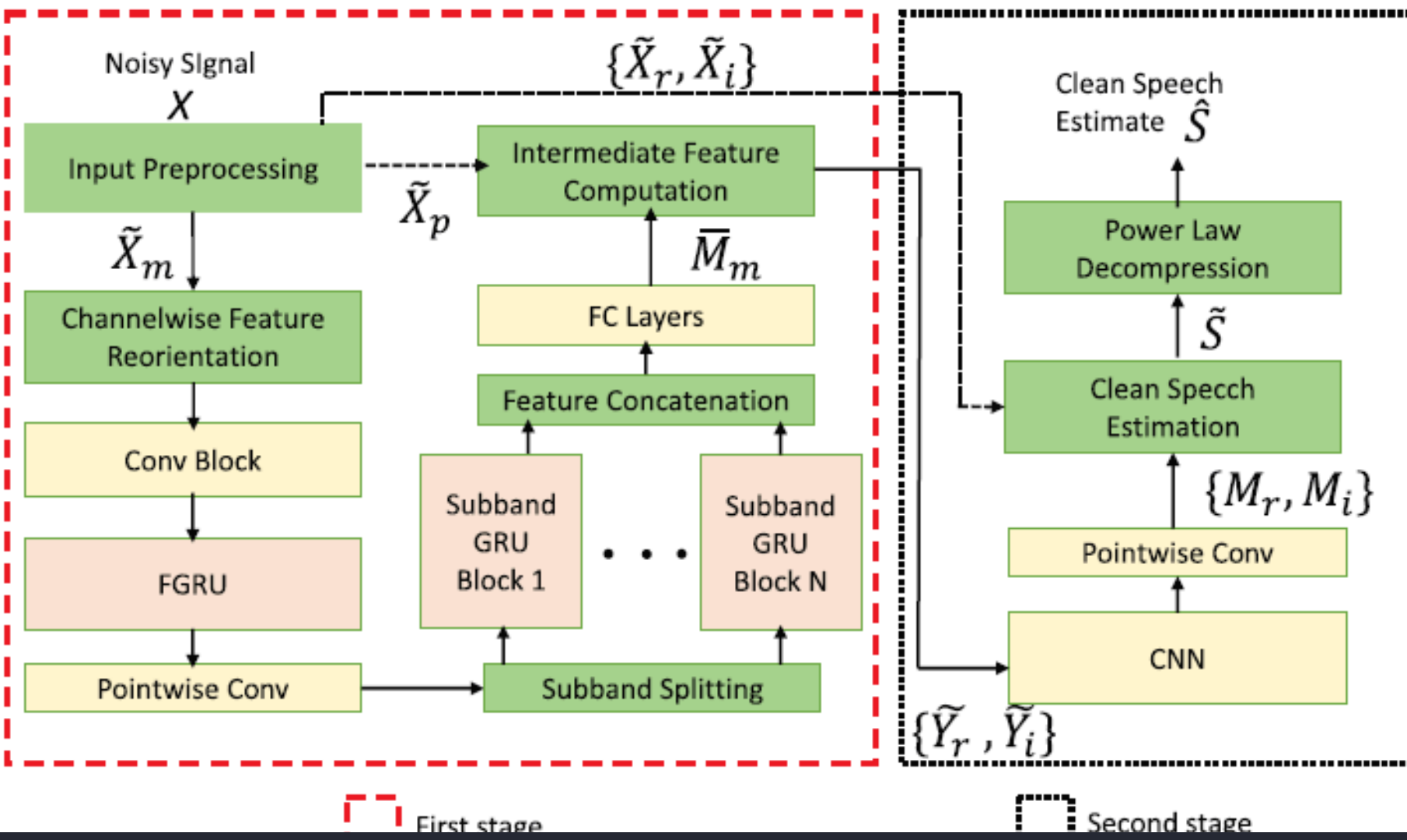


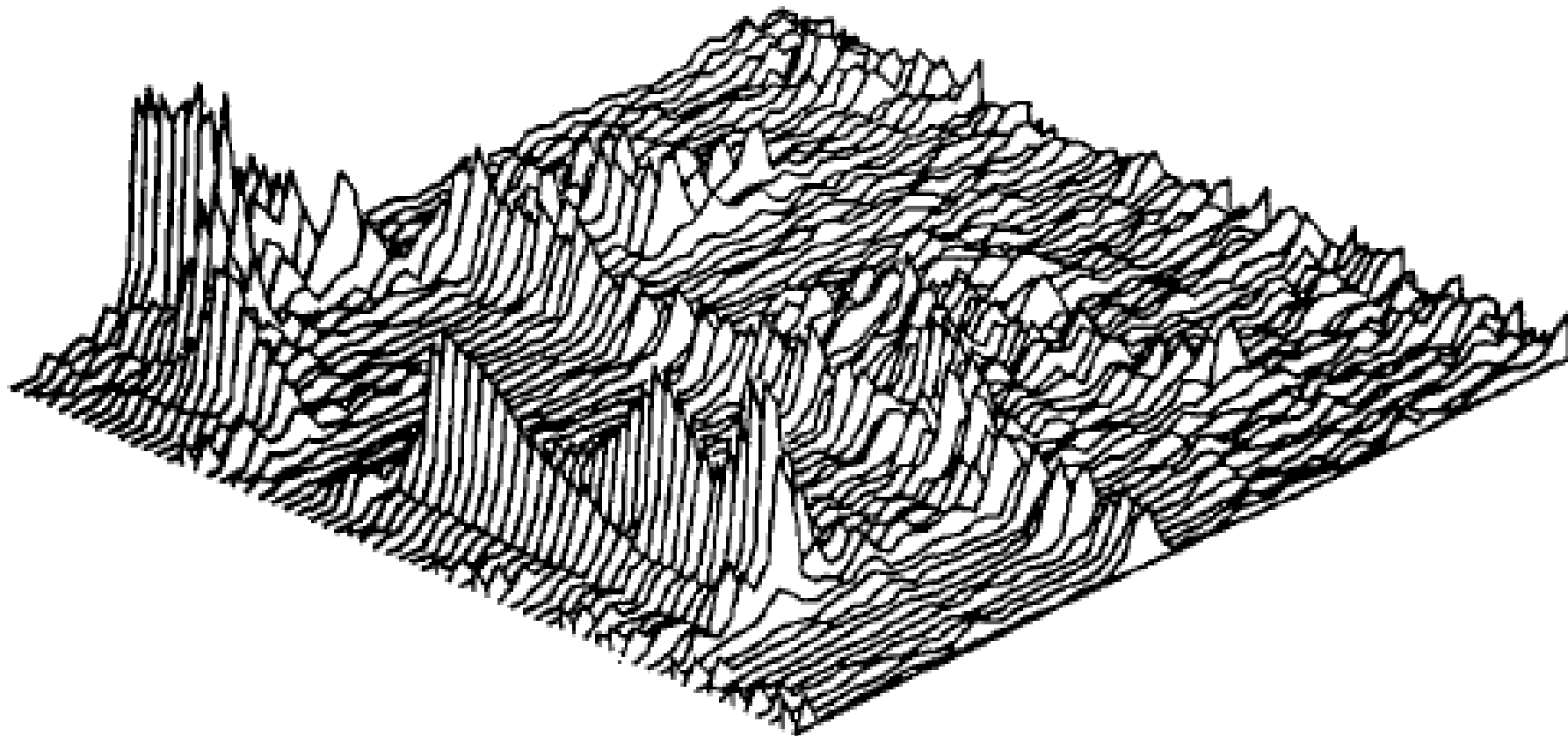


Comparison of Classical and Deep Learning Approaches for Speech Noise Suppression

Aryan Gupta (230150003)

Project Objective

This research compares classical STFT-based speech enhancement with deep learning noise suppression models, demonstrating performance differences through an interactive evaluation framework.



Motivation

The Challenge

Real-world speech signals are frequently degraded by background noise, affecting intelligibility, audio quality, and the performance of automatic speech recognition (ASR) and communication systems.

Two Paradigms

- **Classical signal processing:**
Time-frequency domain operations using noise estimation and filtering
- **Deep learning methods:**
Learned mappings from noisy to clean speech with complex noise pattern modeling

❏ **Research Goal:** Experimentally evaluate the practical differences between these two approaches across multiple performance dimensions.

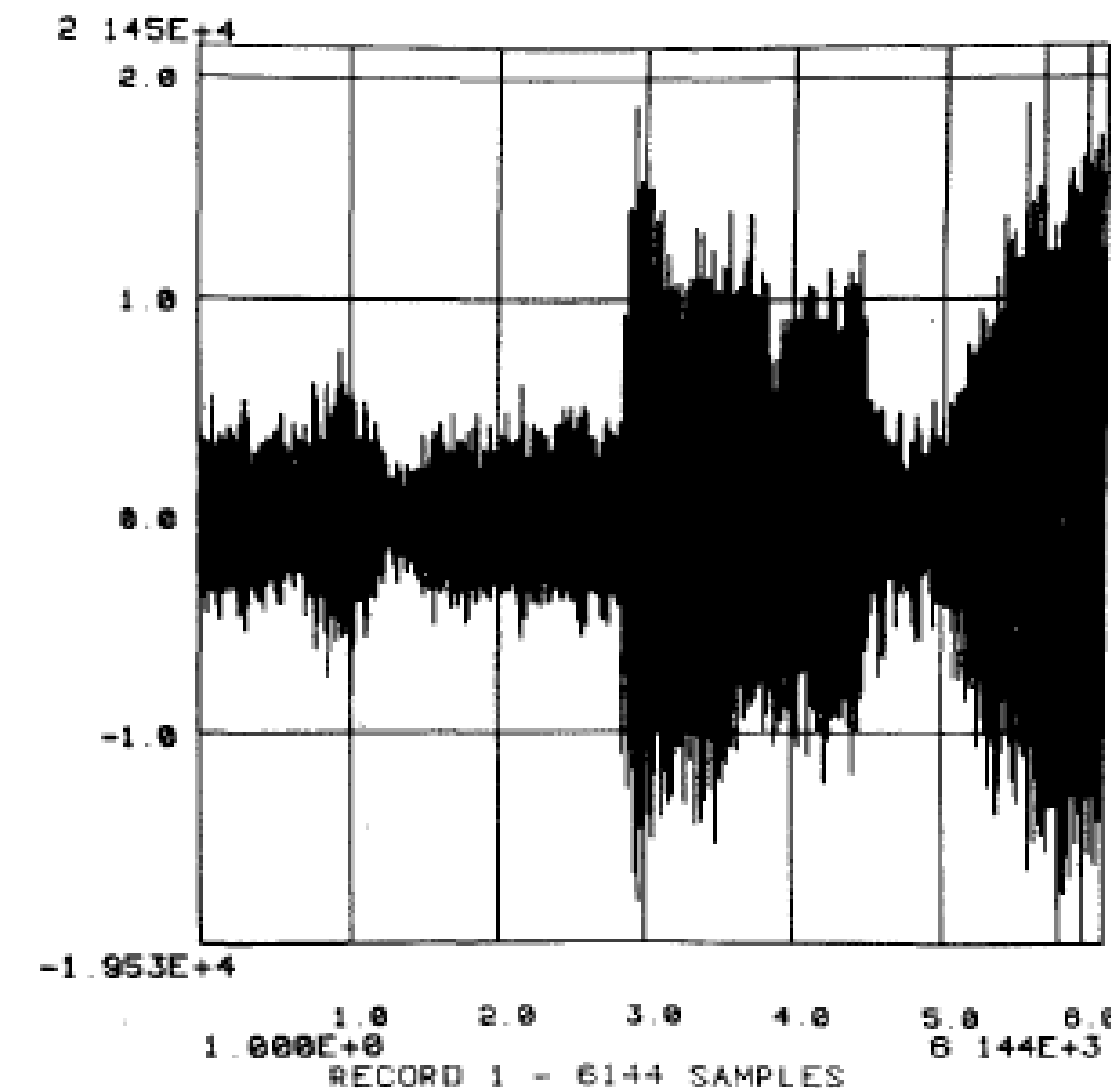


Fig. 4. Time waveform of helicopter speech. "Save your".

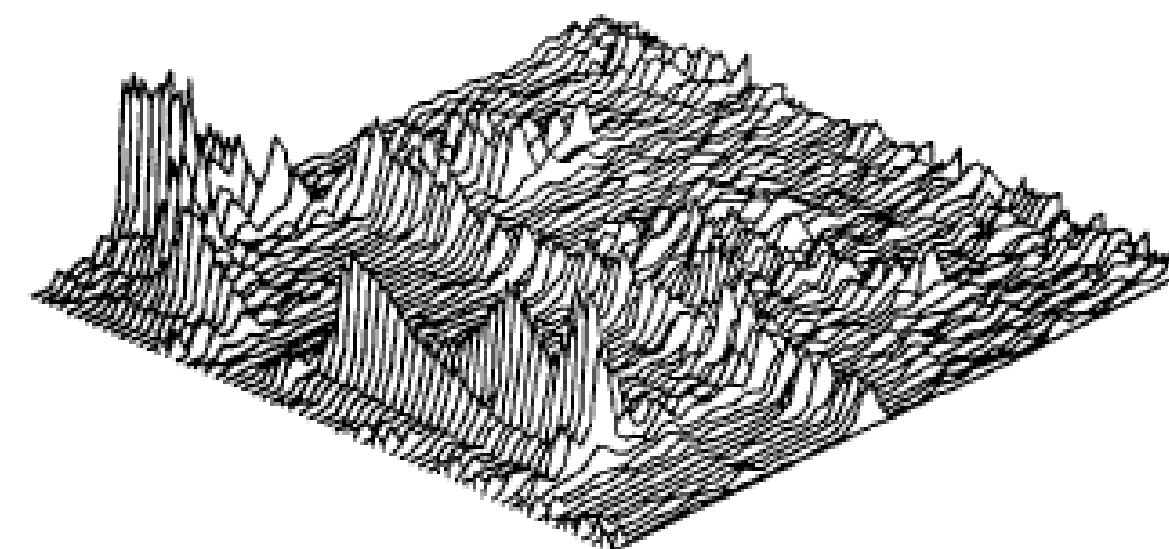


Fig. 6. Short-time spectrum of helicopter speech.

Dataset

Experiments utilize the “Reverberant speech database for training speech dereverberation algorithms and TTS models”, introduced by Cassia Valentini-Botinhao and Junichi Yamagishi.

Dataset Composition

Clean speech recordings from the VCTK corpus combined with noise recordings from the DEMAND dataset.

Training Set

56 speakers (28 male + 28 female) providing diverse vocal characteristics and speaking styles

Test Set

2 speakers reserved for unbiased performance evaluation

Noise Conditions

Multiple real-world noise types at signal-to-noise ratios 0-15 dB in train set and 2.5-17.5 db in test set

Dataset Sample - Noisy Datapoint



Dataset Sample - Corresponding Clean Datapoint



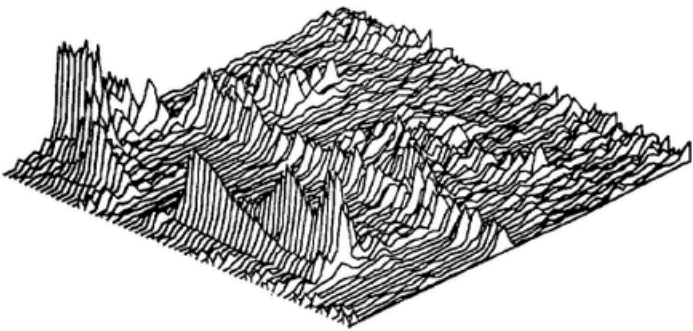


Fig. 6. Short-time spectrum of helicopter speech.

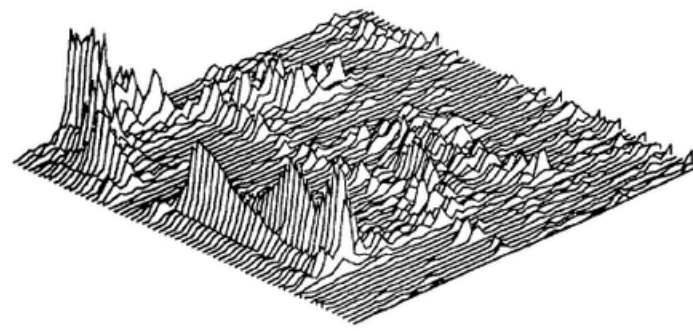


Fig. 7. Short-time spectrum using bias removal and half-wave rectification.

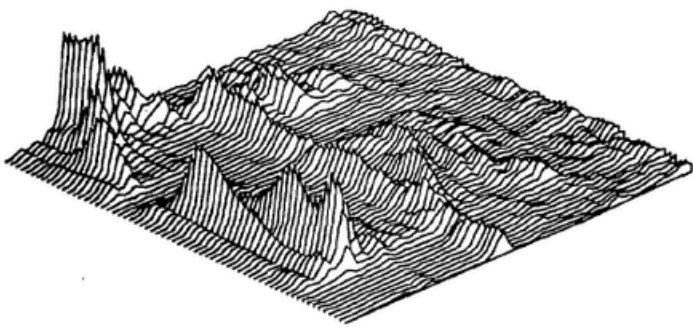


Fig. 8. Short-time spectrum of helicopter speech using three frame averaging.

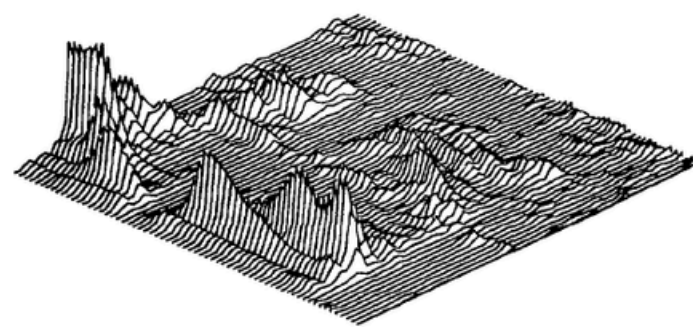


Fig. 9. Short-time spectrum using bias removal and half-wave rectification after three frame averaging.

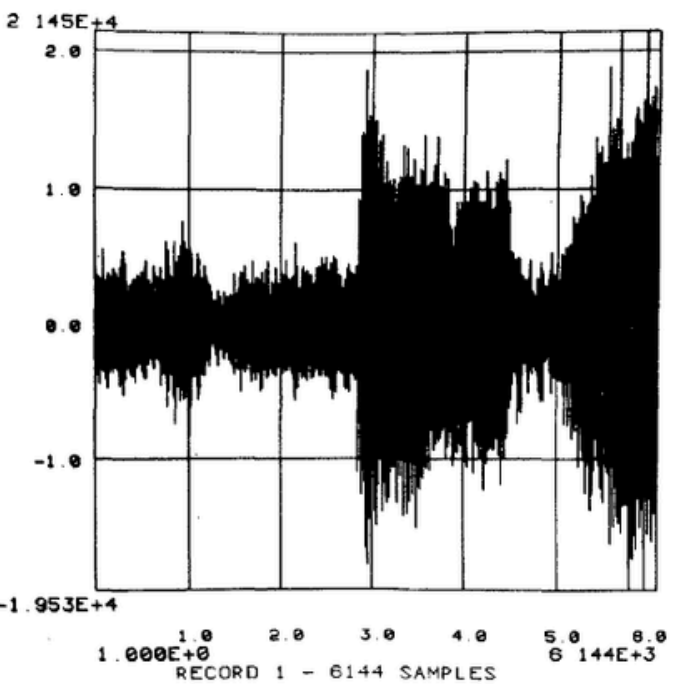


Fig. 4. Time waveform of helicopter speech. "Save your".

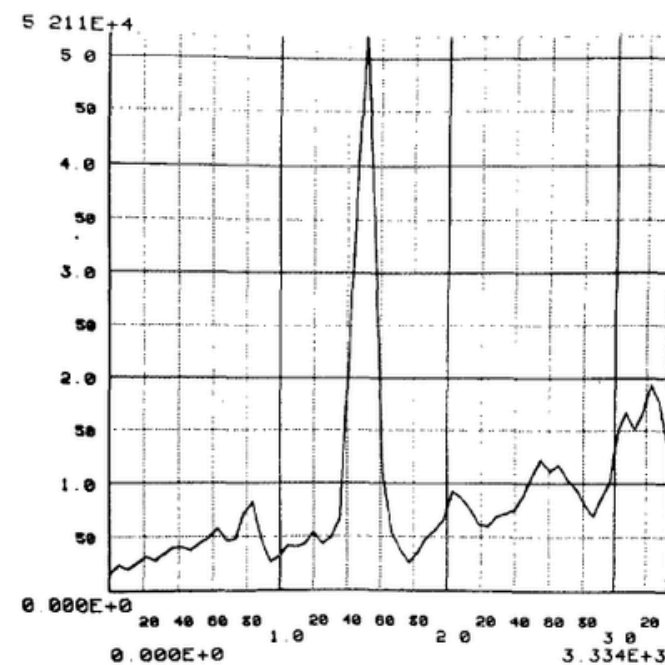


Fig. 5. Average noise magnitude of helicopter noise.

Classical Speech Enhancement Approach

Traditional STFT-Based Processing

Short-Time Fourier Transform (STFT) processing remains the foundation of classical speech enhancement. The well-known spectral subtraction method, proposed by Steven F. Boll, represents a cornerstone technique.

STFT Analysis

Convert noisy speech to frequency domain

Noise Estimation

Estimate noise spectrum characteristics

Spectral Suppression

Suppress noise in magnitude spectrum

Signal Reconstruction

Inverse STFT to recover waveform

Advantages

- Simple, computationally efficient
- Suitable for real-time applications

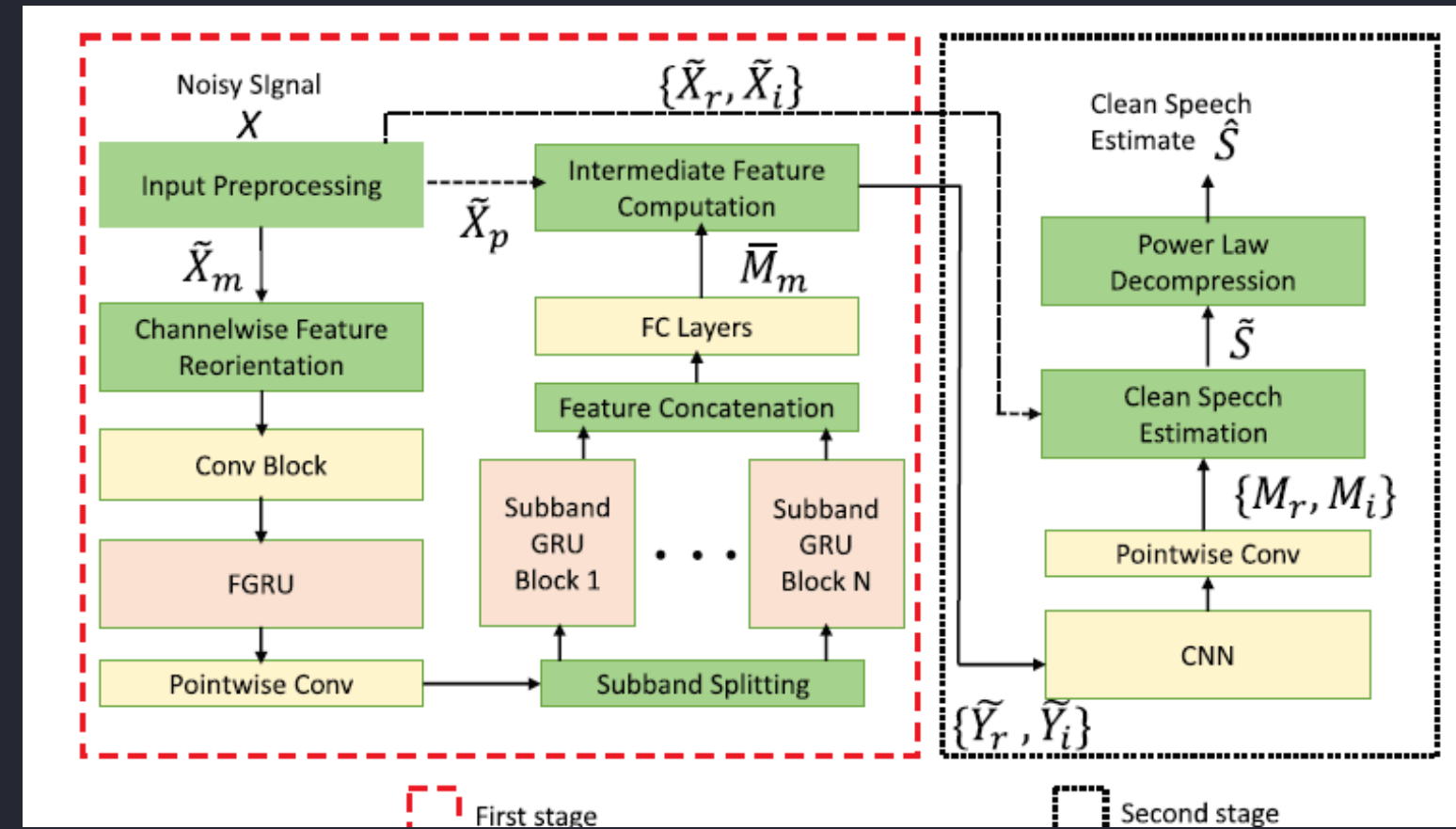
Limitations

- Sensitive to noise estimation errors
- May introduce artifacts

Deep Learning Based Approach

Neural Network Enhancement

Modern speech enhancement leverages neural networks to learn complex mappings from noisy speech to clean representations. This project employs a model inspired by ULCNet operating in the STFT domain.



01

STFT Transformation

Convert noisy speech to time-frequency representation

02

Power-Law Feature Compression

reduce dynamic range and improve learning stability

03

Intermediate Mask Estimation

Neural network predicts spectral mask to identify noise components

04

Complex Mask Refinement

small CNN refines the intermediate estimate and predicts a complex spectral mask to better reconstruct the speech signal

05

Waveform Reconstruction

Inverse STFT recovers enhanced time-domain signal

Improved Performance

Superior noise suppression across diverse noise conditions

Complex Pattern Modeling

Capable of learning intricate noise structures

Environmental Adaptability

Generalizes across different acoustic environments

Evaluation Metrics

Performance assessment employs established objective speech enhancement metrics spanning quality, intelligibility, and computational efficiency dimensions.



Mean Squared Error

MSE between the enhanced speech and the clean speech will give a good estimate of how the model performed.



Average SNR reduction

This will help compare the models. The better model will have more SNR reduction.

Spectral *Subtraction*

Boll's STFT-based denoising — the cornerstone classical method.

- 01 Estimates noise profile from initial ~0.5s audio segment.
- 02 Subtracts noise magnitude spectrum directly from noisy signal.
- 03 Retains original noisy phase for waveform reconstruction.
- 04 Highly efficient and fast, requiring zero model training.
- 05 Fails if speech is present during initial noise estimation.

STFT PIPELINEfig. 01

Waveform $x(t)$ time domain



STFT – $X(k, n)$ windowed FFT



$|X| - |\hat{N}|$ subtract noise profile

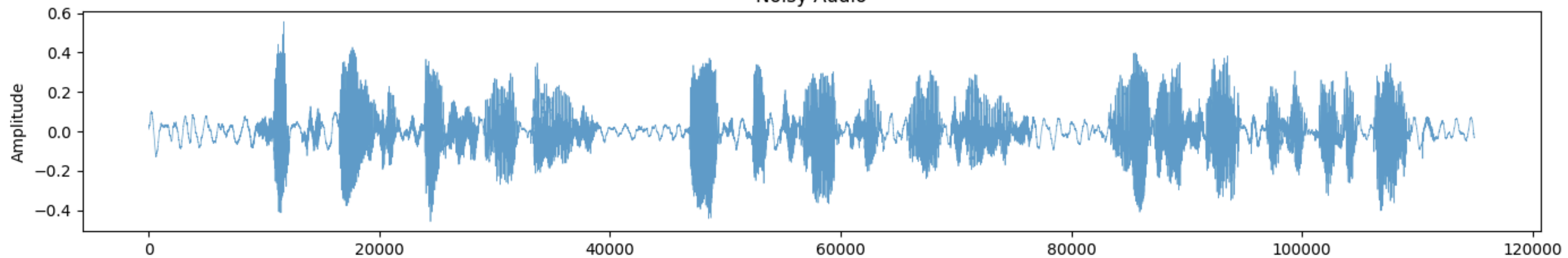


$\max(\cdot, 0)$ half-wave rectify

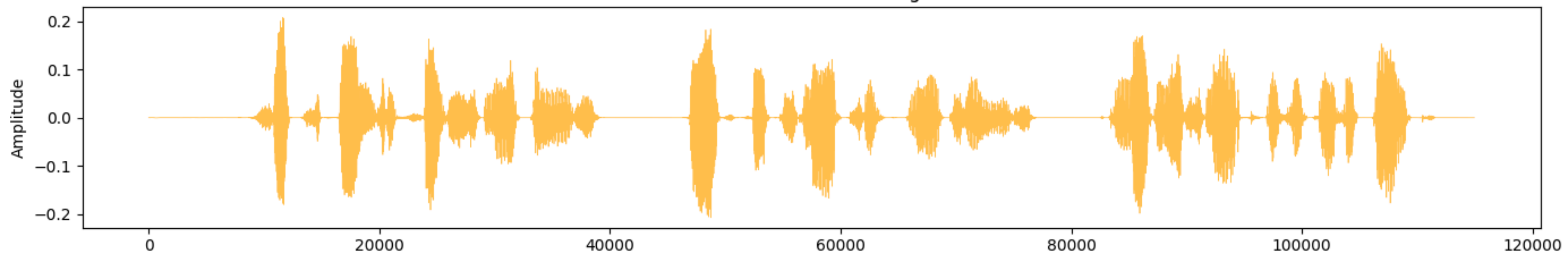


ISTFT $\rightarrow \hat{s}(t)$ enhanced

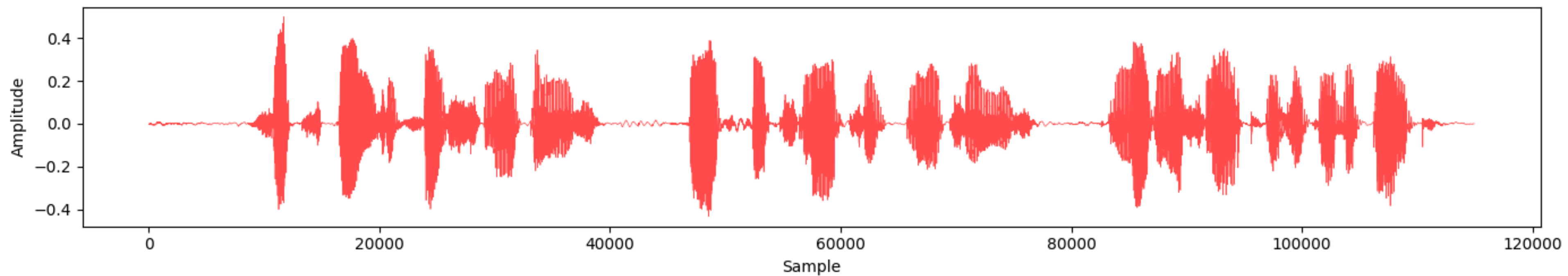
Noisy Audio



Manual STFT Denoising



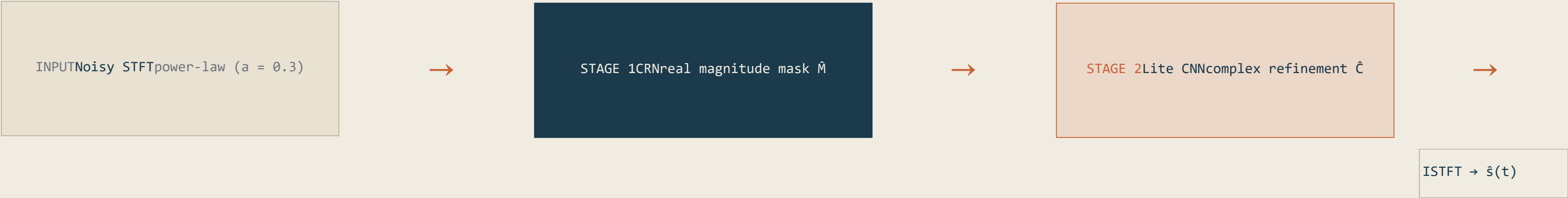
Clean Reference



ULCNet *Overview*

A two-stage learned framework — magnitude masking, then complex refinement.

TWO-STAGE FRAMEWORKfig. 02 – block diagram



- 01

End-to-end learning using shared STFT configurations.
- 02

Utilizes power-law compression (a = 0.3) for robust training.
- 03

Stage 1 estimates a real-valued magnitude mask via CRN.
- 04

Stage 2 refines the complex spectrum using a lightweight CNN.
- 05

Predicts unconstrained masks, allowing selective amplification/attenuation.

Heavy *vs* Light Models

The same task, two operating points — capacity against efficiency.

HEAVY MODEL A

~51Mparams

- CNN based model.
- Bidirectional GRUs for deep temporal context.
- Demands high capacity for complex noise modeling.

TRADEaccuracy ↑ · footprint ↑

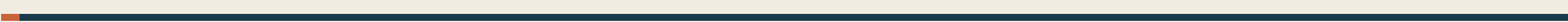
LIGHT MODEL B

~611kparams

- Unidirectional subband GRUs for efficiency.
- 3-stage frequency pooling: 48 → 24 → 12 → 6.
- Significantly reduces footprint while preserving quality.

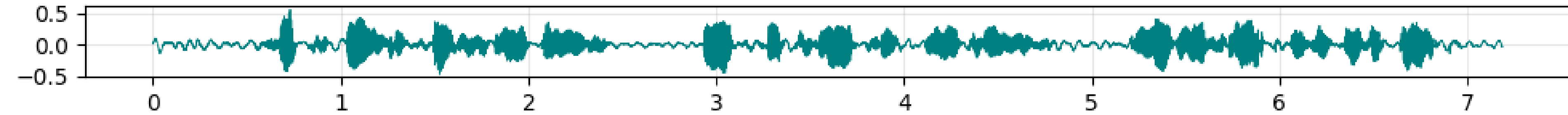
TRADElatency ↓ · footprint ↓

PARAMETER RATIO

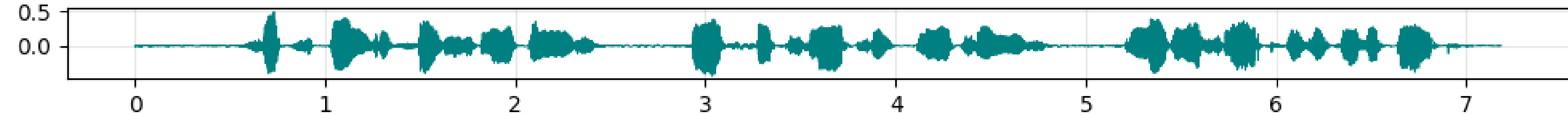


~85× smaller

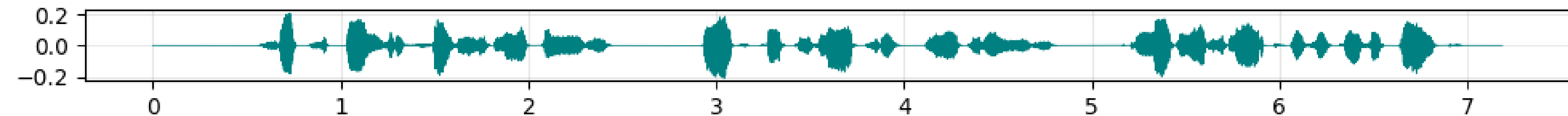
Original Noisy Waveform



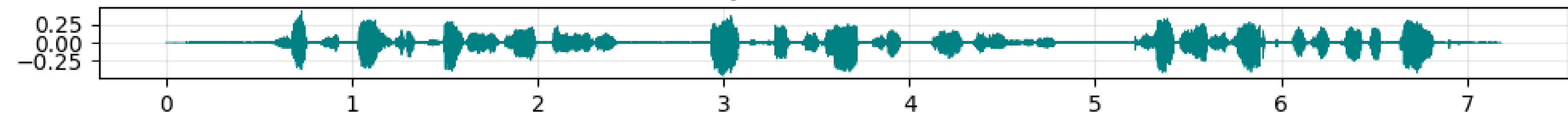
Clean Reference Waveform



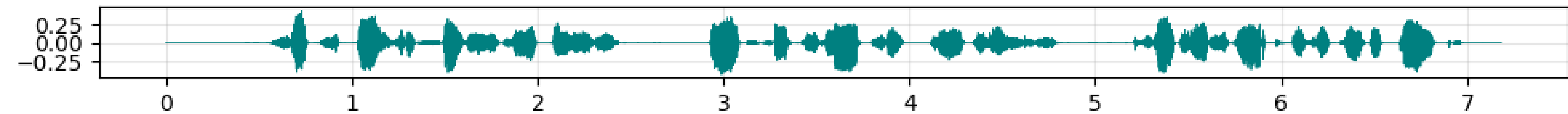
Manual Mode Waveform



Heavy (Vanilla) Waveform



Light (Full) Waveform



Ablation Study

Stage 1 Only

~4k less params

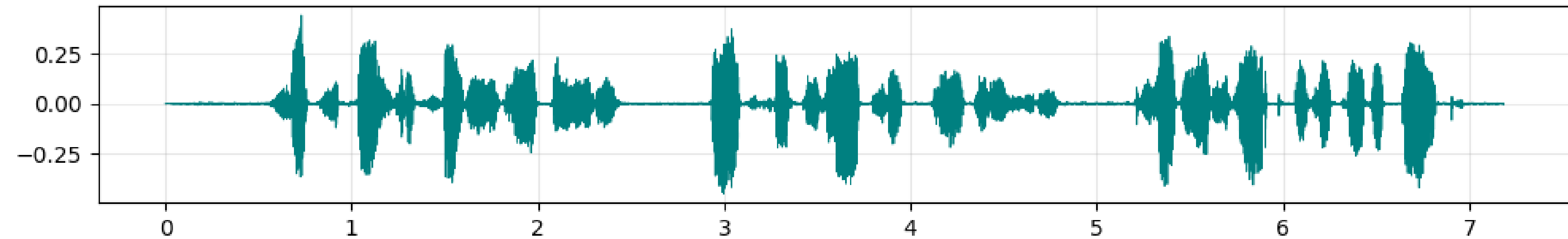
- Uses only stage 1 in the model as it is more important than the second stage.
- The decrease in loss is very low as compared to the increase in number of params

No Compression

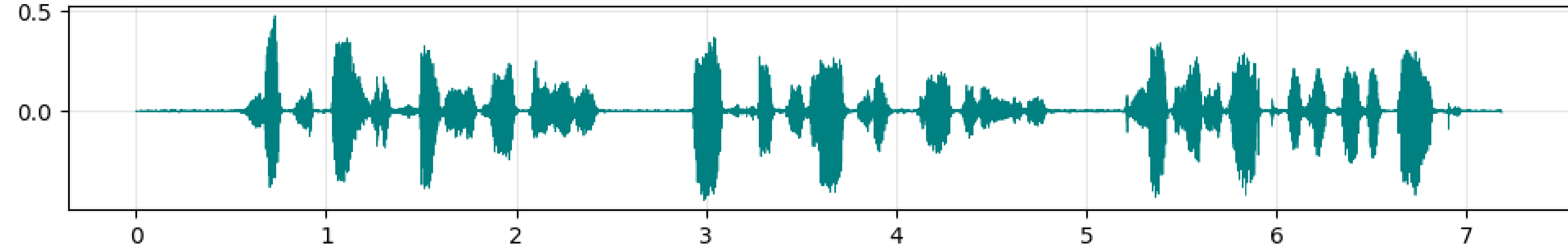
~0 less params

- Sets $\alpha = 0$ for compression.
- Doing this decreases loss significantly.
- This is mainly because it allows the model to work on the data directly rather than going through the compression phase.

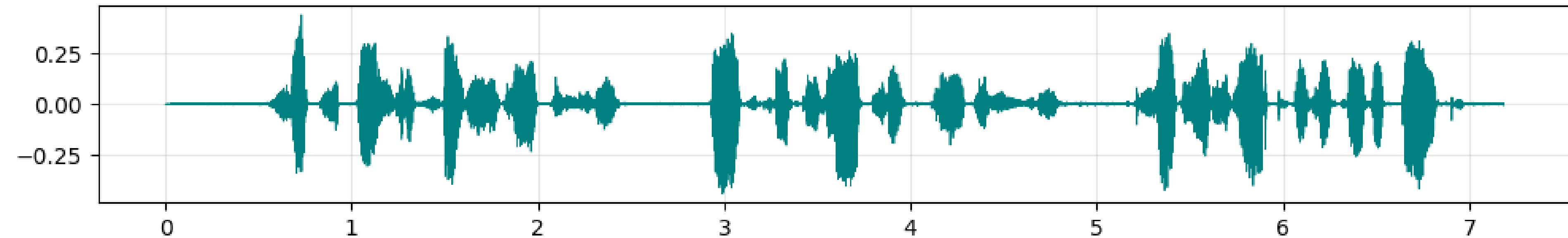
Heavy (Vanilla) Waveform



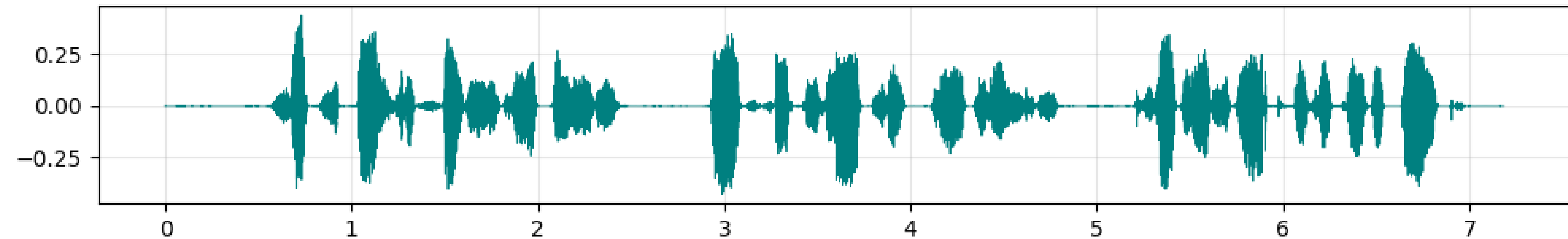
Heavy (Stage1) Waveform



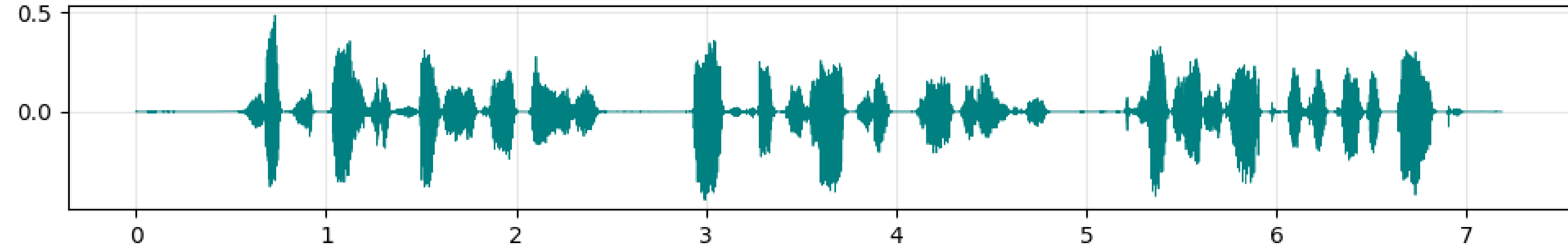
Heavy (No Compression) Waveform



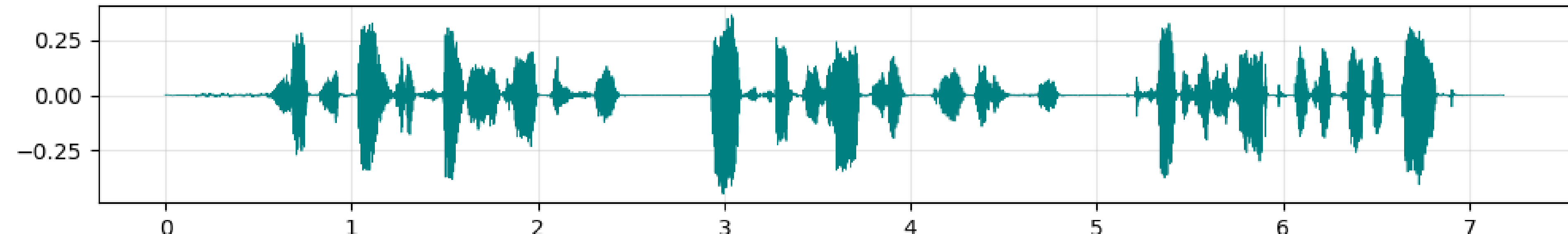
Light (Full) Waveform



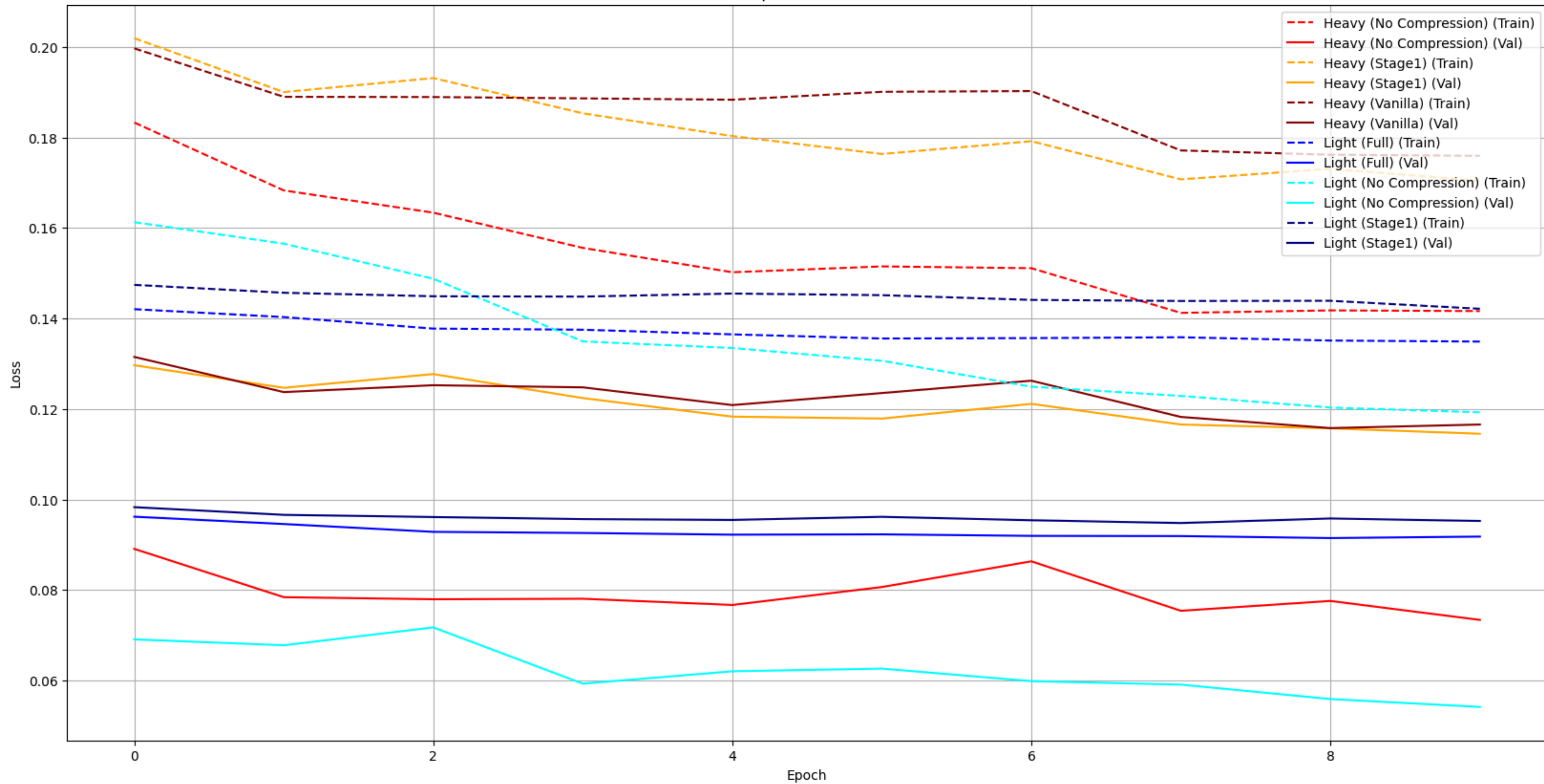
Light (Stage1) Waveform



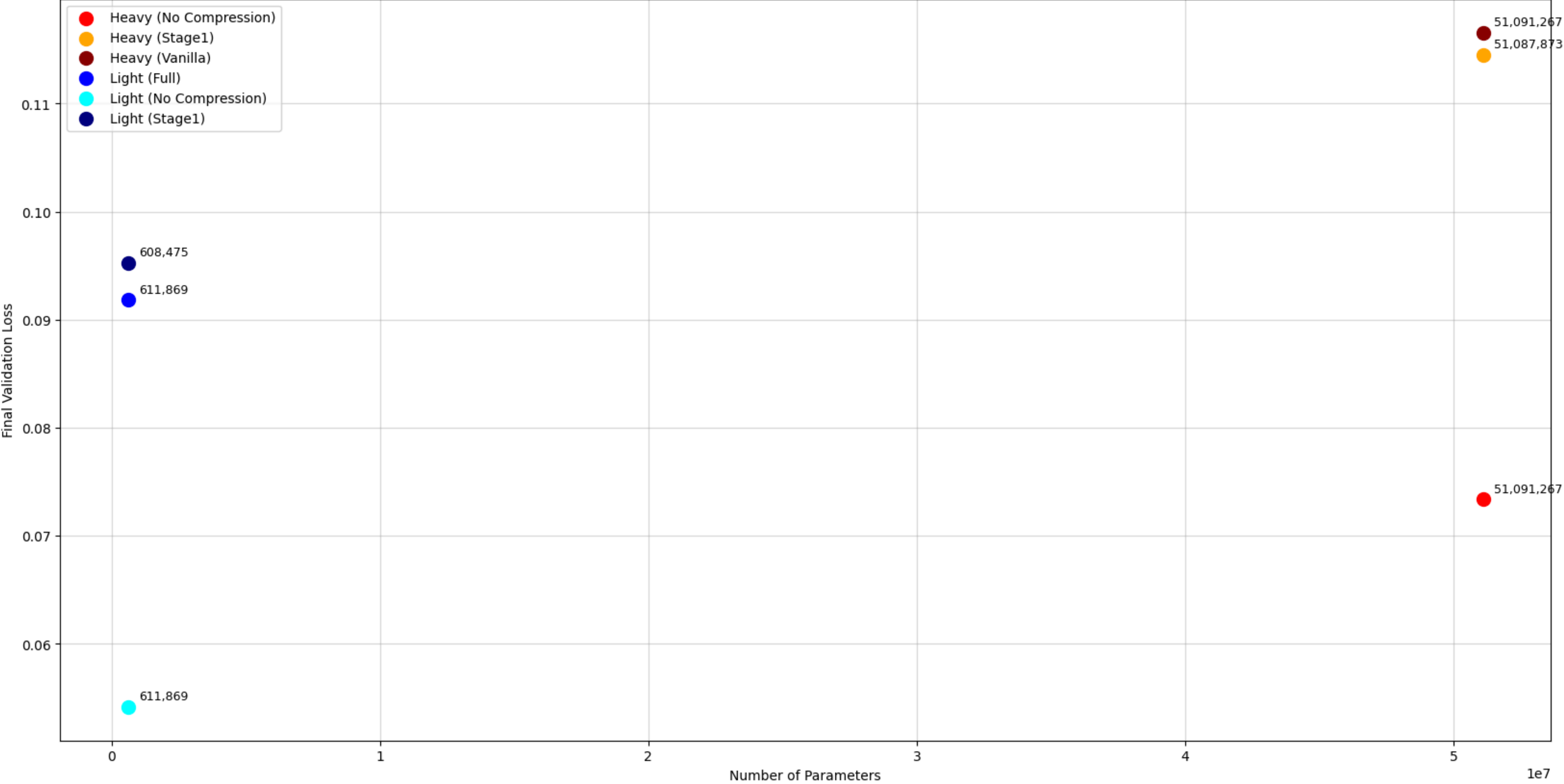
Light (No Compression) Waveform



Epoch vs Loss



Loss vs Parameters (Ablation Study)



Conclusion *& Future Work*

Two paradigms, complementary strengths — the path forward is the blend.

CLASSICAL



Speed & Interpretability

- 01 Classical methods offer extreme speed and high interpretability.
- 02 Future work: integrate adaptive thresholds into classical methods.

LEARNED



Dynamic Adaptation

- 01 Deep learning handles non-stationary and complex noise robustly.
- 02 Future work: deploy lightweight ULCNet on embedded devices.

GOAL

Blend classical efficiency with deep learning adaptability.

Citations



01 S. S. Shetu, S. Chakrabarty, O. Thierngart and E. Mabande, "Ultra Low Complexity Deep Learning Based Noise Suppression," ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Seoul, Korea, Republic of, 2024, pp. 466-470, doi: 10.1109/ICASSP48485.2024.10448353. keywords: {Performance evaluation;Degradation;Computational modeling;Noise reduction;Memory management;Signal processing algorithms;Artificial neural networks;speech enhancement;noise suppression;power law compression;two-stage processing},

02 S. Boll, "Suppression of acoustic noise in speech using spectral subtraction," in IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 27, no. 2, pp. 113-120, April 1979, doi: 10.1109/TASSP.1979.1163209.
keywords: {Acoustic noise;Speech enhancement;Working environment noise;Noise reduction;Speech processing;Speech analysis;Convolution;Narrowband;Speech recognition;Authentication},

03 C. Valentini-Botinhao and J. Yamagishi, "Speech Enhancement of Noisy and Reverberant Speech for Text-to-Speech," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 26, no. 8, pp. 1420-1433, Aug. 2018, doi: 10.1109/TASLP.2018.2828980.
keywords: {Speech;Speech enhancement;Reverberation;Training;Noise measurement;Feature extraction;Remote state estimation;event-based scheduling;generalized closed skew normal distribution}.

Thank You

~Aryan Gupta (230150003)